CS543 • WEB SERVICES

ASSIGNMENT 6 OF 12

Honest Responses: Status Codes, Errors & Validation

Group ID: 17

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Part D — Curl Tests and Submission

Run the endpoints and capture real status codes and bodies

D1. Short theory

curl -i prints the HTTP status line and response headers as well as the body. Use it to verify the status, Location, Content-Type, and Problem Details body. The examples below are templates; replace the endpoint paths and authentication requirements with your actual project settings.

D2. Successful create (happy path)

```
curl -i -X POST http://localhost:8080/api/visualizations \
-H "Content-Type: application/json" \
-H "Accept: application/json" \
-H "Idempotency-Key: run-06-001" \
-d '{"algorithm":"BUBBLE_SORT","input":[5,2,4,1]}'

// Expected contract: 201 Created and Location:
// /api/visualizations/{id}
// Use 202 instead only if your API queues the job.
```

D3. Invalid request

```
curl -i -X POST http://localhost:8080/api/visualizations \
-H "Content-Type: application/json" \
-H "Accept: application/problem+json" \
-H "Idempotency-Key: run-06-002" \
-d '{"algorithm":"UNKNOWN","input":[]}'

// Expected: 422 with errors[] listing algorithm and input.
```

D4. Unknown visualization ID

```
curl -i http://localhost:8080/api/visualizations/not-a-real-id

// Expected: 404 with a Problem Details body.
```

D5. XML negotiation

```
curl -i -H "Accept: application/xml" \
http://localhost:8080/api/algorithms

// Expected: XML if implemented; otherwise 406 Not Acceptable.
```

D6. Rate-limit / unavailable tests

```
// In a test environment, trigger configured rate limiting
// and verify:
// HTTP/1.1 429 Too Many Requests
// Retry-After: 5
// Content-Type: application/problem+json

// For temporary unavailability:
// HTTP/1.1 503 Service Unavailable
// Retry-After: 5
// Content-Type: application/problem+json
```

D7. Regression test for idempotency

```
// Send the exact same POST twice using the same key and body.
// Confirm both responses point to the same logical visualization
// and that the database/history contains only one run.
// A timeout does not prove the server failed: retry with the
// same key and body, not a new key.
```

Notes / implementation caution

Submission checklist: updated service source; updated openapi.yaml with endpoint status codes and Problem Details schema; transcript.txt containing actual curl output for success and failure cases; NOTES.md answering the assignment questions. Add student/team names and roll numbers to the first page.